

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

AUTOMOBILE ENGINEERING

(Mechanical Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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|---|----|
| (a) Define Automobile. Give the typical specifications of an automobile. | 2M |
| (b) What are the functions of piston and crank shaft? | 2M |
| (c) State the advantages and limitations of electronic ignition system. | 2M |
| (d) What is the function of carburettor? | 2M |
| (e) What is power steering? | 2M |
| (f) What is the objective of vehicle suspension? | 2M |
| (g) Give detailed account of a tyre specification. | 2M |
| (h) Write the advantages of ABS braking system over conventional systems. | 2M |
| (i) List the name of automobile battery components. | 2M |
| (j) What is fuel cell? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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|--|----|
| 2 (a) Discuss the methods of vehicle construction in detail. | 6M |
| (b) Write about oil pumps and filters. | 4M |
| OR | |
| 3 (a) Explain with neat sketch the principle of crankcase ventilation. | 4M |
| (b) What is the purpose of lubrication in an automobile engine? Explain any one with neat sketch. | 6M |
| 4 (a) With a neat sketch explain the magneto coil type ignition system. | 5M |
| (b) Describe the working of Electronic fuel injection system with a neat sketch. | 5M |
| OR | |
| 5 (a) Explain working principle of Mechanical fuel pump with neat sketch. | 5M |
| (b) Explain MPFI system for petrol engine with neat sketch. | 5M |
| 6 (a) How is wheel alignment done in automobile? Explain. | 4M |
| (b) Explain with neat sketch the independent rear suspension system. | 6M |
| OR | |
| 7 (a) Explain any one type of steering gear box with neat sketch. | 5M |
| (b) What is a shock absorber? Explain any one in detail. | 5M |
| 8 (a) Draw the neat sketch of construction of tyre and name the parts. | 5M |
| (b) Sketch the arrangement of pneumatic braking system used in automobiles and explain. | 5M |
| OR | |
| 9 (a) Write the wheel types and specifications. Sketch and explain the construction of a disc wheel. | 5M |
| (b) Compare the working of vacuum brakes and hydraulic brakes. | 5M |
| 10 (a) Describe with neat sketch the Active Suspension System. | 5M |
| (b) Explain the construction and lay out of Hybrid Vehicles | 5M |
| OR | |
| 11 (a) With the help of a neat diagram describe Electronic Brake Distribution. | 5M |
| (b) Write short note on Global Positioning System (GPS). | 5M |

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AUTOMOBILE ENGINEERING

(Mechanical Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- | | |
|--|----|
| (a) Explain the need of crank case ventilation. | 2M |
| (b) Write down the function of piston rings. | 2M |
| (c) Define "Multi point injection" of petrol engine. | 2M |
| (d) What is the spark advance and retard mechanism? | 2M |
| (e) Draw a leaf spring. | 2M |
| (f) Write the principle of steering. | 2M |
| (g) What is meant by tandem cylinder? | 2M |
| (h) Why the shock absorbers are used in automobile? | 2M |
| (i) What is the function of a battery in any automobile? | 2M |
| (j) Which type of electrodes is used in a fuel cell? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- | | | |
|-----------|---|-----|
| 2 | Discuss about the chassis and body components in automobile. | 10M |
| OR | | |
| 3 | With the help of a neat sketch explain Splash Lubrication system? Also discuss its advantages and disadvantages. | 10M |
| 4 | Describe in detail about the multipoint fuel injection for S.I. engines. | 10M |
| OR | | |
| 5 | What is the function of a carburettor? Briefly describe the working with a neat sketch. | 10M |
| 6 | Explain the functioning of independent front wheel suspension system with torsion bar in an automobile. | 10M |
| OR | | |
| 7 | Explain the term "Backlash" in Steering Gear. Write the common procedure adopted to adjust the backlash in steering gear. | 10M |
| OR | | |
| 8 | Explain the working principles of Hydraulic braking system with neat sketches. | 10M |
| OR | | |
| 9 | What is ABS and how does ABS helps in improving uncontrolled skidding? | 10M |
| OR | | |
| 10 | What is a fuel cell? With a neat sketch explain its working principle. Write down its limitations. | 10M |
| OR | | |
| 11 | Discuss about the Electronic brake distribution system with suitable sketches. | 10M |
